Grenoble-INP, ENSIMAG
M1 in Applied Mathematics

Internship Report

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Abstract

This report retraces all the work done during a summer internship at the Cagire team. The first part details hyperbolicity in conservation laws and the Godunov method. The second part presents a Julia implementation of the method. Finally the last part is a study of testcases for the Carbuncle phenomenon in the Aerosol software. A new method develloped by the Cagire team showed better handling of these instabilities than standard finite volumes.

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1. Introduction

When simulating supersonic flows, viscosity is often neglected as this results in simple predictable behaviours. These flows can be solved using computationally friendly numerical methods like finite volumes. However in specific cases these simple methods are plagued by numerical artifacts. This internship is focused on understanding these numerical methods and the observed instabilities. A first objective is to implement a first order finite volume method with different Riemann solvers for the Euler equations. Afterwards, we will study test cases that triggers grid-aligned numerical instabilities and add them to the AeroSol software [1]. Finally a curl and divergence preserving method develloped by the team at Cagire will be tested for these instabilities.

This internship was done at the Inria team Cagire in Pau. The team is working on various high performance flow simulationswich include complex physical properties like being multiphasic, turbulent and multiscale. They also work on efficient high order numerical methods for complex geometries.

2. The finite volume method for fluid dynamics

In this first part I will detail the theoretical part of this internship. I will present the equation governing fluid motion, then I will look into their specificities and present numerical methods used for solving them.

2.1. The Euler equations

Describing the dynamics of fluids by each particle composing the medium is too difficult. Instead physicists use macroscopic quantities that result from taking the mean of microscopic values over many particles.

For compressible fluids these macroscopic quantities are :

- the **density** $\rho(\boldsymbol{x}, t)$ in $\mathrm{kg} \cdot \mathrm{m}^{-3}$ resulting from taking the mean mass per volume.
- the **velocity vector** $\boldsymbol{v}(\boldsymbol{x}, t) = \begin{pmatrix} u & v & w \end{pmatrix}^T$ in $\mathrm{m} \cdot \mathrm{s}^{-1}$ resulting from taking the mean velocity of each particle per volume.
- the **pressure** $p(\boldsymbol{x}, t)$ in $\mathrm{Pa} = \mathrm{N} \cdot \mathrm{m}^{-2}$ resulting from the mean magnitude of force per area.

These are called the *primitive* variables, but in order to compute them we use the *conserved variables* :

- the **density** $\rho(\boldsymbol{x}, t)$
- the **momentum vector** $\boldsymbol{m}(\boldsymbol{x}, t) = \rho \boldsymbol{v}$
- the **total energy per unit volume** $E(\boldsymbol{x}, t) = \rho\left(\frac{1}{2}\boldsymbol{v} \cdot \boldsymbol{v} + e\right)$

with e the **specific internal energy** that is determined using thermodynamical laws (EOS). We are gonna use the EOS for ideal gases given by $e(\rho, p) = \frac{p}{(\gamma-1)\rho}$.

Using the fundamental laws of conservation of mass, Newton second law and the conservation of Energy we get the Euler equations governing inviscid compressible fluids on the *conserved variables* :

$$\begin{cases} \rho_t + \boldsymbol{\nabla} \cdot (\rho \boldsymbol{v}) = 0 \\ (\rho \boldsymbol{v})_t + \nabla \cdot (\rho \boldsymbol{v} \otimes \boldsymbol{v} + pI) = 0 \\ E_t + \nabla \cdot [(E+p)\boldsymbol{v}] = 0 \end{cases} \tag{2.1}$$

The system can be rewritten in conservative form :

$$\boldsymbol{u}_t + \boldsymbol{F}(\boldsymbol{u})_x + \boldsymbol{G}(\boldsymbol{u})_y + \boldsymbol{H}(\boldsymbol{u})_z = 0 \tag{2.2}$$

with $\boldsymbol{u} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ E \end{pmatrix}$, $\boldsymbol{F}(\boldsymbol{u}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho uw \\ (E+p)u \end{pmatrix}$, $\boldsymbol{G}(\boldsymbol{u}) = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho vw \\ (E+p)v \end{pmatrix}$ and $\boldsymbol{H}(\boldsymbol{u}) = \begin{pmatrix} \rho w \\ \rho uw \\ \rho vw \\ \rho w^2 + p \\ (E+p)w \end{pmatrix}$

$\boldsymbol{u}$ is the column vector of conserved variables and $\boldsymbol{F}, \boldsymbol{G}, \boldsymbol{H}$ are the *flux vectors* in the x, y, z directions respectively. This is a system of differential equations and thus assumes smooth solutions (partial derivatives exist). In order to handle discontinuous solutions we will need to rewrite it in integral form.

2.1.1. Submodels

There exists a lot of derived systems obtained from simplifications on the Euler equations. These can be used to analyze each aspect of the full system individually and reduce computation when approximations can be made.

2.1.1.1. Linear Waves and Linear Advection

Let us consider a linearization of the Euler equation around a state at rest $\boldsymbol{v} = \begin{pmatrix} \rho_0 \\ 0 \\ p_0 \end{pmatrix}$ in which we introduce a small disturbance : $\boldsymbol{v}' = \begin{pmatrix} \rho' \\ u' \\ p' \end{pmatrix}$.

When simplifying derivatives of constants and higher order terms, the one dimensional Euler equation becomes

$$\begin{cases} \rho'_t + \rho_0 u'_x = 0 \\ u'_t + \left(\frac{1}{\rho_0}\right) p'_x = 0 \\ p'_t + \rho_0 a_0^2 u'_x = 0 \end{cases} \tag{2.3}$$

with $a_0 = \sqrt{\frac{\gamma p_0}{\rho_0}}$ the speed of sound.

As ρ' only appear in the first equation, it is fully determined by integrating $\rho_0 u'_x$ over time.

The remaining 2×2 system is called the wave system. It's specificity is that it combines two underlying linear waves travelling in opposite direction. This drops the non linearity in the full euler system, as the coefficient $\left(\frac{1}{\rho_0}\right)$ and $\rho_0 a_0^2$ are constants independent from the state vector $\boldsymbol{v}'$.

2.1.1.2. Burgers' Equation

The burgers' equation on the other hand is a scalar non linear equation as the coefficient in front of u_x depends on the state u :

$$u_t + u u_x = 0 \tag{2.4}$$

This equation is the simplest non-linear conservation law, and as such is useful to study non-linearity.

2.1.1.3. Isentropic Euler Equations

Finally, a more complete system before jumping to the full Euler system is obtained when assuming a constant entropy, $\mathrm{d}S = 0$. This leads to pressure being fully described by the density field, $p(\rho) = \kappa \rho^\gamma$, with κ and γ two constants. We can thus drop the last equation of the euler system.

For $\gamma = 2$ and $\kappa = \frac{g}{2}$ we obtain the same equations as in the Shallow Water Equations.

2.2. Hyperbolicity

A key aspect of all of these systems is hyperbolicity. For smooth solutions, the chain rule lets us rewrite the one dimensional conservative system (2.2) in *quasilinear form*

$$\boldsymbol{u}_t + \boldsymbol{A}(\boldsymbol{u}) \boldsymbol{u}_x = 0 \tag{2.5}$$

where $\boldsymbol{A} = \nabla_{\boldsymbol{u}} F(\boldsymbol{u})$ is the Jacobian matrix of the flux.

Definition 2.2.1: The system (2.5) is **hyperbolic** if $\boldsymbol{A}$ has real eigenvalues $\lambda_1 \leq \dots \leq \lambda_m$ and a complete set of linearly independent right eigenvectors $\boldsymbol{r}_1, ..., \boldsymbol{r}_m$. It is **strictly hyperbolic** if in addition the eigenvalues are distinct.

Thus there exists an invertible matrix formed by the right eigenvectors $P(\boldsymbol{u}) = (\boldsymbol{r}_1, ..., \boldsymbol{r}_m)$ and a diagonal matrix with the diagonal elements being the eigenvalues $D(\boldsymbol{u}) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$ such that $A(\boldsymbol{u}) = P(\boldsymbol{u}) D(\boldsymbol{u}) P(\boldsymbol{u})^{-1}$.

This is especially useful for **linear** hyperbolic conservation laws as P and D do not depend on $\boldsymbol{u}$. In that case, we can use the change of variable

$$\boldsymbol{v} = P^{-1} \boldsymbol{u} \tag{2.6}$$

These are called *characteristic variables* and using them, the quasilinear form can be rewritten as a system of m independent advection equations :

$$\frac{\partial v_i}{\partial t} + \lambda_i \frac{\partial v_i}{\partial x} = 0 \tag{2.7}$$

Where λ_i are the diagonal entries of D, i.e. the eigenvalues of $\nabla_{\boldsymbol{u}} F(\boldsymbol{u})$ and are called the *characteristic speeds.* In the more general non-linear case, this can't be done as the eigenvalues depend on the state so no such decoupling of the system exists.

2.2.1. Relation to the classification of second order equations

The *hyperbolic* name comes from the classification of second order scalar equation into elliptic, parabolic and hyperbolic.

A second order equation of the form $au_{xx} + bu_{xy} + cu_{yy} = 0$ is classified using the discriminant $\Delta = b^2 - 4ac$. If $\Delta > 0$ the equation is hyperbolic, if $\Delta = 0$ the equation is parabolic and if $\Delta < 0$ the equation is elliptic.

By setting $\boldsymbol{w} = \binom{u_x}{u_y}$ and using the condition $(u_x)_t = (u_t)_x$, we get a first order system :

$$\begin{gathered}\boldsymbol{w}_x + A\boldsymbol{w}_y = 0 \\ \text{with } A = \begin{pmatrix} \frac{b}{a} & \frac{c}{a} \\ -1 & 0 \end{pmatrix}\end{gathered} \tag{2.8}$$

The characteristic polynomial of A is given by $\lambda^2 - \frac{b}{a}\lambda + \frac{c}{a} = 0$ and it has real roots when $\frac{b^2-4ac}{a^2} > 0$, so the hyperbolicity condition is the same for both definitions.

The classification of the equations translate in their behaviour. Hyperbolic equations are wave like with a finite propagation speed. This isn't the case for parabolic equations whose diffusive behaviour affects the whole domain without any speed limit. Finally elliptic problems do not have any real characteristic direction and thus no timelike direction.

For example, steady state heat conductivity, modelized by Laplace's equation $-\Delta u = 0$ is elliptic and adding a time dependency $u_t - \Delta u = 0$ makes it parabolic. Linear advection, $u_t + cu_x = 0$ on the other hand is Hyperbolic and so are the equations presented in Section 2.1.1. The full Euler equation however are Hyperbolic only under certain condition and can become elliptic, for example in a steady subsonic compressible model. Classification becomes even more complex when using the Navier Stokes equations that add a diffusive term.

2.2.2. Characteristic curves

Definition 2.2.2.1 (characteristic curves): Characteristics are curve in the $x - t$ plane where the PDE reduces to an ODE in time.

These curve are a very useful tool to visualize the wave like propagation of hyperbolic systems.

Let us first study the linear scalar advection $u_t + cu_x = 0$ and let us consider a curve $x = x(t)$. We can thus write u as a function of t $u(t) = u(x(t), t)$. Differentiating u over t along the curve $x(t)$ gives :

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\partial u}{\partial t} + \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\partial u}{\partial x} \tag{2.9}$$

This is similar to the LHS of the PDE. In the case where the curve satisfies $\frac{\mathrm{d}x}{\mathrm{d}t} = c$ we have $\frac{\mathrm{d}u}{\mathrm{d}t} = 0$. This means that $u(t) = u_0$ is constant along the curve. Similarly to isolines for spatial visualization, the characteristics show us the path of constant values over time. From $\frac{\mathrm{d}x}{\mathrm{d}t} = c$ we obtain $x(t) = ct + x_0$, on the $x - t$ plane these give lines with a slope $\frac{1}{c}$.

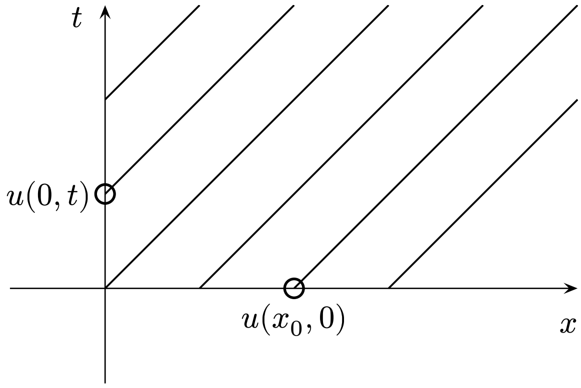

Figure 1: Characteristic curves in the (x,t) plane for linear advection.

The values of u on the rays are given either by the initial condition or the boundary condition.

Linear hyperbolic systems can be decomposed into independent scalar advection equations on the characteristic variables as detailed in (2.7). Each of these variables are then getting transported by their correspond speed given by the eigenvalues.

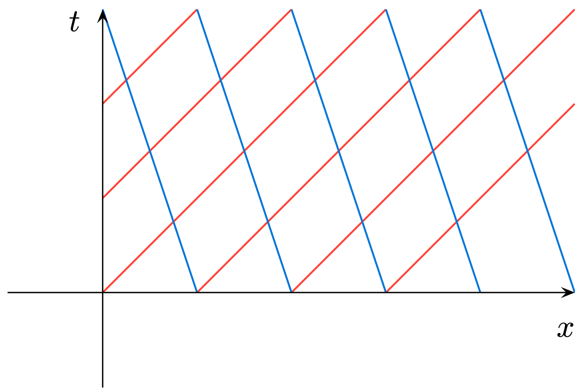

Figure 2: Characteristic curves for a 2×2 hyperbolic system.

This can be seen when simulating the wave system in 1D and watching the characteristic variables. Introducing a small difference in the initial condition, one can see one wave travelling to the left and the other to the right when viewing each characteristic variable.

Let us now consider a scalar hyperbolic conservation law $u_t + f(u)_x = 0$. For f differentiable, we can write it in quasilinear form $u_t + \lambda(u)u_x = 0$ with $\lambda(u) = f'(u)$. If we then consider characteristic curves satisfying

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \lambda(u), \quad x(0) = x_0 \tag{2.10}$$

then the local derivative of $u(t) = u(x(t),t)$ along $x(t)$ gives

$$\frac{\mathrm{d}u}{\mathrm{d}t} = u_t + \lambda(u)u_x = 0 \tag{2.11}$$

i.e. u is constant on this curve whose value is given by following the characteristic back in time, crossing either the initial value or a boundary condition. The slope can then be evaluated as $\lambda(u_0)$ and the curve is given by $x(t) = x_0 + \lambda(u_0(x_0))t$. This means that two rays can have different slopes which leads to crossings, where a single point has multiple values and the model breaks at time t_c.

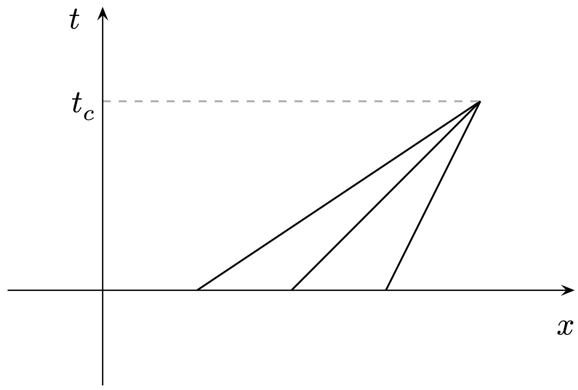

Figure 3: Crossing of characteristic curves.

In order to keep an inviscid model we have to allow discontinuities in the computed solutions. This can be done by searching for weak solution using the integral form of the conservation law.

2.2.3. Weak solutions

Crossing characteristics force us to accept solutions that are not differentiable in the classical sense. We only require $\boldsymbol{u}$ to be locally integrable, this is possible thanks to distribution theory.

Definition 2.2.3.1 (weak solution): $\boldsymbol{u}$ is a *weak solution* of $\boldsymbol{u}_t + \boldsymbol{F}(\boldsymbol{u})_x = 0$ if for every test function $\varphi \in C_0^1(\mathbb{R} \times \mathbb{R}^+)$

$$\int_0^\infty \int_{\mathbb{R}} (\boldsymbol{u}\varphi_t + \boldsymbol{F}(\boldsymbol{u})\varphi_x)\,\mathrm{d}x\,\mathrm{d}t + \int_{\mathbb{R}} \boldsymbol{u}(x,0)\varphi(x,0)\,\mathrm{d}x = 0. \tag{2.12}$$

The weak formulation admits discontinuous data, which the classical theory could not. The simplest such data is a single jump, and the resulting initial value problem is central to everything that follows.

Definition 2.2.3.2 (Riemann problem): The *Riemann problem* for a conservation law is the initial value problem

$$\begin{cases} \boldsymbol{u}_t + \boldsymbol{F}(\boldsymbol{u})_x = 0 \\ \boldsymbol{u}(x,0) = \begin{cases} \boldsymbol{u}_L & \text{if } x<0 \\ \boldsymbol{u}_R & \text{if } x>0 \end{cases} \end{cases} \tag{2.13}$$

with two constant states $\boldsymbol{u}_L$ and $\boldsymbol{u}_R$.

This problem is the main building block of the Godunov method.

Now consider a discontinuity between two smooth states to travel at speed S and evaluate the integral form on a control volume encompassing this discontinuity. Taking its limit yields the Rankine-Hugoniot condition, also called the jump condition (see [2], p. 70):

$$\boldsymbol{F}(\boldsymbol{u}_R) - \boldsymbol{F}(\boldsymbol{u}_L) = S(\boldsymbol{u}_R - \boldsymbol{u}_L) \tag{2.14}$$

2.2.4. Shocks and Rarefactions

When characteristics cross, we now approximate it by a discontinuity called a **shock**. This happens when a characteristic on the left is faster than on the right, $\lambda(u_l) > \lambda(u_r)$ for a scalar law. For fluids, a shock represents a rapid transition layer from one data state to another. This layer is usually of the order of the mean free path of the molecules, and thus can be approximated by a discontinuity. Determining the speed S of this wave is solving the Riemann problem (2.13). This can be solved using the Rankine-Hugoniot conditions (2.14).

Example 2.2.4.1: For Burgers' equation, the Rankine-Hugoniot condition gives $S = \frac{1}{2}(u_r + u_l)$, that is the shock travels at the mean velocity between the left and right states.

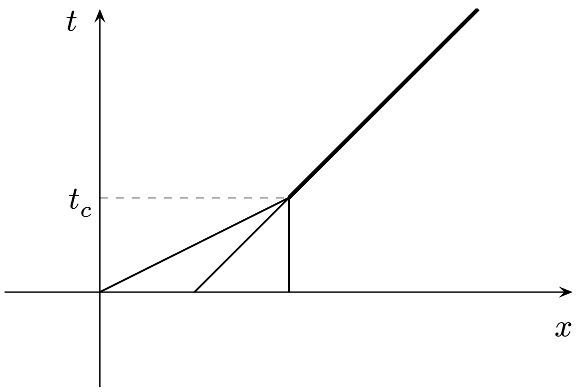

Figure 4: Creation of a shock

Another situation can arise when we consider discontinuities, when the characteristics do not move into, but outwards from a discontinuity. For a scalar law, this happens when $\lambda(u_l) < \lambda(u_r)$ and the problem is again given by the Riemann problem (2.13).

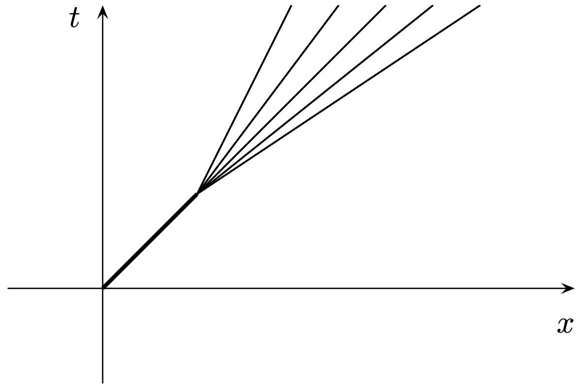

Figure 5: A discontinuity diverging into multiple characteristics, leaving a region undetermined

A solution would be to continue representing the wave by a discontinuity creating an expansion shock. This would respect the Rankine-Hugoniot condition (2.14) but is not an acceptable solution. If we try to determine the value on the rays by tracing it backwards in time, we end up at the discontinuity which doesn't carry any information. We therefore impose an additional admissibility criterion for discontinuities.

Definition 2.2.4.1 (Lax entropy condition): A discontinuity of speed S in the λ_i field is admissible if

$$\lambda_i(\boldsymbol{u}_L) > S > \lambda_i(\boldsymbol{u}_R). \tag{2.15}$$

This means that for a discontinuity to be admissible, the characteristics should go into the shock, which isn't the case for the expansion shock.

The correct physical solution called a **rarefaction fan** is constructed using self-similarity, that is it doesn't change depending of the scale of the variables.

For a scalar law, setting $u(x,t) = \tilde{u}(\xi)$, with $\xi = \frac{x}{t}$ gives

$$(\lambda(\tilde{u}(\xi)) - \xi)\tilde{u}'(\xi) = 0 \tag{2.16}$$

so at each point either $\tilde{u}' = 0$, giving a constant state, or

$$\lambda(\tilde{u}(\xi)) = \xi. \tag{2.17}$$

and for λ invertible, we obtain

$$\tilde{u}(\xi) = \lambda^{-1}(\xi) \tag{2.18}$$

A discontinuous jump would have to satisfy (2.14), and the only such jump is the expansion shock rejected above, hence the solution must be continuous. This forces $\tilde{u}(\xi) \to u_l$ as $\xi \to \lambda(u_l)$ and $\tilde{u}(\xi) \to u_r$ as $\xi \to \lambda(u_r)$. The fan is therefore bounded by the two characteristics of slopes $\lambda(u_l)$ and $\lambda(u_r)$, called its *head* and *tail* respectively, and (2.18) gives the solution between them.

Example 2.2.4.2: For Burgers' equation $\lambda(u) = u$ is the identity, so the fan is simply $u = x/t$, with head u_l and tail u_r.

2.2.5. The Riemann problem for non-linear hyperbolic systems

Now that we looked into non-linear scalar quasi-linear forms, let us now look how systems behave on Riemann problems. Instead of having only one characteristics, systems have several, ordered by their respective speed $\lambda_1(\boldsymbol{u}) \leq ... \leq \lambda_i(\boldsymbol{u}) \leq ... \leq \lambda_m(\boldsymbol{u})$. Assuming strict hyperbolicity, the eigenvalues are distinct and each family is associated with a single eigenvector direction. Each family then produces one wave separating two constant states.

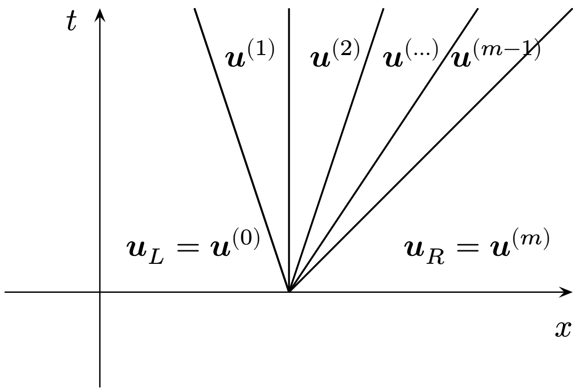

Figure 6: Example solution to the Riemann problem for a hyperbolic system of conservation laws

In order to find the solution to (2.13), we have to find each of these $m - 1$ intermediate states as we have $m + 1$ regions and we know the left and the right one. In total that represents $m(m - 1)$ unknowns as each unknown intermediate state has m components.

Let us look at a specific characteristic family i. It separates two states $\boldsymbol{u}^{(i-1)}$ on the left and $\boldsymbol{u}^{(i)}$ on the right. We have $\boldsymbol{u}^{(0)} = u_L$ and $\boldsymbol{u}^{(m)} = u_R$. As seen for the scalar case, if the associated characteristic speed on the left is greater than the right one $\lambda_i\left(\boldsymbol{u}^{(i-1)}\right) > \lambda_i\left(\boldsymbol{u}^{(i)}\right)$, we get a shock. The Rankine-Hugoniot conditions give us m equations. If we know one of the two data states, this still leaves one unknown as we have to also determine the shock speed S.

If instead the left characteristic is smaller than the right one, we get a rarefaction fan. Substituting $\boldsymbol{u}(x, t) = \tilde{\boldsymbol{u}}(\xi)$ into the quasilinear form (2.5) gives

$$\boldsymbol{A}(\tilde{\boldsymbol{u}})\tilde{\boldsymbol{u}}'(\xi) = \xi\tilde{\boldsymbol{u}}'(\xi) \tag{2.19}$$

so inside the fan $\tilde{\boldsymbol{u}}'$ is a right eigenvector of $\boldsymbol{A}$ with eigenvalue ξ. This yields the scalar relation $\lambda_i(\tilde{\boldsymbol{u}}(\xi)) = \xi$ for one of the m eigenvalues i of A fixed throughout the rarefaction fan. As for the scalar case, this gives us the solution inside the fan when λ_i is invertible : $\tilde{\boldsymbol{u}}(\xi) = \lambda_i^{-1}(\xi)$. But we still want to find a relationship between the left and right data states $\boldsymbol{u}^{(i-1)}$ and $\boldsymbol{u}^{(i)}$. Let us use the other information provided by (2.19), $\tilde{\boldsymbol{u}}'(\xi)$ is an eigenvector of $\boldsymbol{A}$ associated with the eigenvalue λ_i as such we have :

$$\tilde{\boldsymbol{u}}'(\xi) \in \operatorname{span}(\boldsymbol{r}_i(\tilde{\boldsymbol{u}})), \tag{2.20}$$

with $\boldsymbol{r}_i$ the right eigenvector associated with λ_i. This means there exist $\alpha(\xi)$ such that $\tilde{\boldsymbol{u}}'(\xi) = \alpha(\xi)\boldsymbol{r}_i(\tilde{\boldsymbol{u}})$. Componentwise we get $\frac{\tilde{\boldsymbol{u}}'^{(j)}(\xi)}{\boldsymbol{r}_i^{(j)}(\tilde{\boldsymbol{u}}(\xi))} = \alpha(\xi)$ and multiplying by $\mathrm{d}\xi$ gives

$$\frac{\tilde{\boldsymbol{u}}'^{(1)}\,\mathrm{d}\xi}{\boldsymbol{r}_i^{(1)}(\tilde{\boldsymbol{u}})} = \frac{\tilde{\boldsymbol{u}}'^{(2)}\,\mathrm{d}\xi}{\boldsymbol{r}_i^{(2)}(\tilde{\boldsymbol{u}})} = \ldots = \frac{\tilde{\boldsymbol{u}}'^{(m)}\,\mathrm{d}\xi}{\boldsymbol{r}_i^{(m)}(\tilde{\boldsymbol{u}})} = \alpha\,\mathrm{d}\xi \tag{2.21}$$

We have $\tilde{\boldsymbol{u}}'^{(j)}\,\mathrm{d}\xi = \mathrm{d}\tilde{\boldsymbol{u}}^{(j)}$ and dropping $\alpha\,\mathrm{d}\xi$ gives $m-1$ relations involving only the state variables called the *Generalized Riemann Invariants* :

$$\frac{\mathrm{d}\tilde{\boldsymbol{u}}^{(1)}}{\boldsymbol{r}_i^{(1)}(\tilde{\boldsymbol{u}})} = \frac{\mathrm{d}\tilde{\boldsymbol{u}}^{(2)}}{\boldsymbol{r}_i^{(2)}(\tilde{\boldsymbol{u}})} = \ldots = \frac{\mathrm{d}\tilde{\boldsymbol{u}}^{(m)}}{\boldsymbol{r}_i^{(m)}(\tilde{\boldsymbol{u}})} \tag{2.22}$$

Each equality is an ordinary differential equation in two of the state variables. These $m-1$ relations give us a way to associate $m-1$ variables from the left state to the right state, leaving one undetermined as in the shock case.

Finally in systems we can have one more case, $\lambda_i\big(\boldsymbol{u}^{(i-1)}\big) = \lambda_i\big(\boldsymbol{u}^{(i)}\big)$ with $\boldsymbol{u}^{(i-1)} \neq \boldsymbol{u}^{(i)}$ we get a discontinuity that is neither a shock nor a rarefaction. This can't happen in the scalar case when we assume the flux to be convex or concave. The characteristic speed is given if we know either state across the discontinuity, $S = \lambda_i\big(\boldsymbol{u}^{(i)}\big) = \lambda_i\big(\boldsymbol{u}^{(i-1)}\big)$. The Rankine Hugoniot conditions apply as we have a discontinuity and so do the Generalized Riemann Invariants. For such waves, we have enough equations to determine one state from the other unlike shocks and rarefactions.

2.2.6. Characteristic Fields and wave types

Now we have seen 3 types of wave encountered in one dimensional non-linear hyperbolic conservation laws. We have *contact* waves, *shock* wave and *rarefaction* waves.

The one dimensional equations studied during this internship have the property to only have such waves. Furthermore, we can categorize the waves by the type of the field defined by each eigenvalue $\lambda(\boldsymbol{u})$ in state space.

Definition 2.2.6.1 (Linearly degenerate fields): A λ_i-field is called *linearly degenerate* if

$$\nabla_{\boldsymbol{u}}\lambda_i(\boldsymbol{u}) \cdot \boldsymbol{r}_i(\boldsymbol{u}) = 0, \quad \forall \boldsymbol{u} \in \mathbb{R}^m \tag{2.23}$$

This means that $\lambda_i(\boldsymbol{u})$ is constant along the direction $\boldsymbol{r}_i$ which is the only one the state varies across the i-th wave. Such fields can only produce contact waves.

Definition 2.2.6.2 (Genuinely non-linear fields): A λ_i-field is called *genuinely non-linear* if

$$\nabla_{\boldsymbol{u}}\lambda_i(\boldsymbol{u}) \cdot \boldsymbol{r}_i(\boldsymbol{u}) \neq 0, \quad \forall \boldsymbol{u} \in \mathbb{R}^m \tag{2.24}$$

This means that $\lambda_i(\boldsymbol{u})$ is strictly monotone along the direction span$(\boldsymbol{r}_i)$, which again is the only one the state varies across the i-th wave. Such fields generate either shocks or rarefactions and cannot change their wave type. This also ensures that λ_i is invertible and thus make the solutions for rarefactions possible.

One can also have fields that are neither linearly degenerate nor genuinely non-linear creating composite waves. These are encountered in more complicated settings like MHD and do not appear in the classical Euler equations. For the scalar case, this reduces to assuming that the flux functions are either linear, concave or convex.

2.2.7. Analysis of the Euler equations

The conservative formulation of the Euler equations (2.2) can be rewritten in quasi-linear form using the Jacobian of the flux $\nabla_{\boldsymbol{u}}\boldsymbol{F}(\boldsymbol{u})$ as

$$\boldsymbol{u}_t + \nabla_{\boldsymbol{u}}\boldsymbol{F}(\boldsymbol{u})\boldsymbol{u}_x = 0 \tag{2.25}$$

The eigenvalues of the Jacobian are

$$\lambda_1 = u - a, \quad \lambda_2 = u \text{ and } \lambda_3 = u + a \tag{2.26}$$

which are real and distinct as long as $a \neq 0$. The Euler equation are thus Hyperbolic. The associated right eigenvectors are

$$\boldsymbol{r}_1 = \begin{pmatrix} 1 \\ u - a \\ H - ua \end{pmatrix}, \quad \boldsymbol{r}_2 = \begin{pmatrix} 1 \\ u \\ \frac{1}{2}u^2 \end{pmatrix} \text{ and } \boldsymbol{r}_3 = \begin{pmatrix} 1 \\ u + a \\ H + ua \end{pmatrix} \tag{2.27}$$

where $H = \frac{E+p}{\rho}$ the total specific enthalpy and $a = \sqrt{\frac{\gamma p}{\rho}}$ the sound speed. For the computations see [2] pp. 87-90.

Computing $\nabla\lambda_{i(\boldsymbol{u})} \cdot \boldsymbol{r}_i$, we get that the λ_1 and λ_3 fields are genuinely non-linear. The waves generated by these two fields are called *accoustic waves* and ρ, u and p either change smoothly for a rarefaction or discontinuously for shocks. On the other hand the λ_2 field is linearly degenerate. It's generated wave is called the *entropy wave* and only the density jumps discontinuously while the pressure and the particle velocity stay constant. As $a > 0$, we have that $\lambda_1 < \lambda_2 < \lambda_3$ so the structure of the solution to the Riemann problem will consist of a contact wave surrounded by two non-linear waves (shocks and/or rarefactions).

Until now we only looked at one dimensional systems. If we introduce additional spatial dimensions, the eigenvalue u_n (velocity normal to the considered direction) is not of multiplicity one any more and we lose strict hyperbolicity. This linearly degenerate field now carries multiple waves, the entropy wave wich is a density jump and the shear waves wich are tangential velocity jumps.

Finally we will also consider the isentropic Euler equations that drops the energy equation removing the entropy wave. We are left with only the accoustic and shear waves and it stays strictly hyperbolic up to two dimensions.

2.3. Numerical methods for hyperbolic systems

Now we will look at the Godunov method, a finite-volumes method designed to solve hyperbolic systems of partial differential equations such as the Euler system.

2.3.1. The finite volumes method

The finite volume method results from considering the integral form of conservation laws:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \boldsymbol{u}\,\mathrm{d}v + \int_\Sigma \mathcal{H}\cdot n\,\mathrm{d}\sigma = 0 \tag{2.28}$$

where u is the conserved quantity vector, V the control volume, Σ the boundary of V, $\mathcal{H} = (\boldsymbol{F}, \boldsymbol{G}, \boldsymbol{H})$ the flux tensor and n the outward pointing normal to Σ. This is then enforced on each cell of a given discretization of the domain.

Consider a conservation law :

$$\boldsymbol{u}_t + \sum_{i=1}^{m} F_i(\boldsymbol{u})_{x_i} = 0 \tag{2.29}$$

the integral form gives :

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \boldsymbol{u}\,\mathrm{d}v = -\sum_{\Sigma_k\in\Sigma}\int_{\Sigma_k}\sum_{i=1}^{m} F_i(\boldsymbol{u})n_i\,\mathrm{d}\sigma \tag{2.30}$$

Where (Σ_i) is a set of faces (or edges in 2D) such that $\cup\,\Sigma_i = \Sigma$ and $\Sigma_i \cap \Sigma_j = \varnothing$.

It is left to choose a numerical approximation for the integral, a timestepping method and most importantly a way to compute the fluxes.

2.3.2. Godunov's method

In 1959, Godunov developed a first-order finite volume method for non-linear hyperbolic conservation laws like the Euler system [3].

The method considers piecewise constant approximation using the cell averages $\overline{\boldsymbol{u}} = \frac{1}{|V|}\int_V \boldsymbol{u}\,\mathrm{d}V$. To compute the fluxes it uses an exact one dimensional Riemann solver, that is an algorithm that can find $\boldsymbol{u}^*(x,t)$, the solution of the Riemann problem (2.13). The left and right states $\boldsymbol{u}_L$ and $\boldsymbol{u}_R$ are the cell averages of the two neighboring cells separated by the face and the numerical flux is given by $\boldsymbol{F}(\boldsymbol{u}^*(0,t))$, evaluating the solution on the curve $\frac{x}{t} = 0$.

Conceptually we are letting the fluid flow for some small timestep before taking the average of the cell. As we add and subtract the same amount, the flux, between two cells, this method is conservative.

To determine the timestep we use the criteria that the fastest wave should not travel more than one cell in one timestep. Supposing that no wave acceleration takes place as a consequence of wave interaction, for structured meshes this translates to :

$$\Delta t \leq \frac{\Delta x_i}{\max(|\lambda|)} \tag{2.31}$$

with λ the eigenvalues of the jacobian of the flux in the direction x_i. Then using a coefficient $0 < C_{\text{cfl}} < 1$ we obtain :

$$\Delta t = C_{\text{cfl}} \frac{\Delta x_i}{\max(|\lambda|)} \tag{2.32}$$

Godunov only presented this method for structured meshes, where each direction has its own flux function F_i. Under the condition that the system is *rotationally invariant*, which the Euler equations are, this approach is generalized to unstructured meshes. Using the rotation matrix $T(\boldsymbol{n})$ that transforms vector components aligning them with the normal coordinate axis and leaves the scalar component unchanged, we get :

$$\sum_i F_i(\boldsymbol{u}) n_i = T(\boldsymbol{n})^{-1} F_1(T(\boldsymbol{n})\boldsymbol{u}) \tag{2.33}$$

This in turn makes it possible to approximate the face flux as $\int_{\Sigma_k} \sum_i F_i(\boldsymbol{u}) n_i \,\mathrm{d}\sigma \approx |\Sigma_k| T^{-1} \hat{F}$, where $T^{-1} = T(\boldsymbol{n})^{-1}$. $\hat{F} = F_1(\boldsymbol{u}^*_{\mathbf{rot}})$ is the numerical flux obtained using $\boldsymbol{u}^*_{\mathbf{rot}}$ the solution of a one dimensional Riemann solver in the rotated coordinates.

For a forward in time approximation, the update would then read :

$$\overline{\boldsymbol{u}}_i^{n+1} = \overline{\boldsymbol{u}}_i^n - \frac{\Delta t}{|V_i|} \sum_{\Sigma_k \in \Sigma} |\Sigma_k| T^{-1} \hat{F} \tag{2.34}$$

2.3.3. Exact Riemann Solvers

A key aspect to Godunov's method is computing the solution to the Riemann problem (2.13). These can be solved exactly using an algorithm called an *exact Riemann solver*.

2.3.3.1. Riemann solvers for linear systems

When the system is linear we have independent transport equations on the characteristic variables v_i each advected with a speed λ_i. The solution to the individual Riemann problems are given by

$$v_i(x,t) = \begin{cases} v_{L,i} & \text{if } \frac{x}{t} < \lambda_i \\ v_{R,i} & \text{else} \end{cases} \tag{2.35}$$

The full solution is then just a superposition of all these individual solution, $\boldsymbol{u}(x,t) = \sum v_i(x,t)\boldsymbol{r}_i$. Finally the flux is computed by $F(\boldsymbol{u}(0,t))$.

This can be simplified, as we are only interested in $\boldsymbol{u}(0,t)$ we can only determine $v_i(0,t)$. If $\lambda_i > 0$ the flow goes to the right then $v_i = v_{L,i}$ on the contrary if $\lambda_i < 0$, $v_i = v_{R,i}$ and finally $\boldsymbol{u}(0,t) = \sum_i v_i \boldsymbol{r}_i$.

Example 2.3.3.1.1 (The wave equations):

The wave equations is a one-dimensional linear system :

$$\begin{cases} u_t + \frac{1}{\rho} p_x = 0 \\ p_t + \kappa u_x = 0 \end{cases} \tag{2.36}$$

It can be rewritten in conservative form :

$$\boldsymbol{u}_t + F(\boldsymbol{u})_x = 0 \tag{2.37}$$

with $\boldsymbol{u} = \binom{u}{p}$ and $F(\boldsymbol{u}) = \binom{\frac{1}{\rho}p}{\kappa u}$

Diagonalizing the Jacobian gives :

$$\nabla F(\boldsymbol{u}) = \begin{pmatrix} 0 & \frac{1}{\rho} \\ \kappa & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ \rho c & -\rho c \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2\rho c} \\ \frac{1}{2} & -\frac{1}{2\rho c} \end{pmatrix} \tag{2.38}$$

with $c = \sqrt{\frac{\kappa}{\rho}}$. We have

$$\boldsymbol{u}_t + \nabla F(\boldsymbol{u}) \boldsymbol{u}_x = 0 \tag{2.39}$$

The characteristic variables are :

$$\boldsymbol{v} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2\rho c} \\ \frac{1}{2} & -\frac{1}{2\rho c} \end{pmatrix} \boldsymbol{u} = \begin{pmatrix} \frac{u}{2} + \frac{p}{2\rho c} \\ \frac{u}{2} - \frac{p}{2\rho c} \end{pmatrix} \tag{2.40}$$

and we can write :

$$\boldsymbol{v}_t + \begin{pmatrix} c & 0 \\ 0 & -c \end{pmatrix} \boldsymbol{v}_x = 0 \tag{2.41}$$

We obtain two independent transport equations for each characteristic variable. One is advected to the right with characteristic speed c and the other is advected to the left with a characteristic speed $-c$.

This gives $v_1(0,t) = v_{1,L}$ and $v_2 = v_{2,R}$ and

$$\begin{aligned} \boldsymbol{u}(0,t) &= v_{1,L} \boldsymbol{r}_1 + v_{2,R} \boldsymbol{r}_2 \\ &= \left(\frac{u_L}{2} + \frac{p_L}{2\rho c} \right) \begin{pmatrix} 1 \\ \rho c \end{pmatrix} + \left(\frac{u_R}{2} - \frac{p_R}{2\rho c} \right) \begin{pmatrix} 1 \\ -\rho c \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} u_L + u_R + \frac{p_L - p_R}{\rho c} \\ (u_L - u_R)\rho c + (p_L + p_R) \end{pmatrix} \end{aligned} \tag{2.42}$$

Finlly the Godunov flux is

$$F(u(0,t)) = \frac{1}{2}\begin{pmatrix} (u_L - u_R)\rho c + \frac{p_L + p_R}{\rho} \\ \kappa\Big(u_L + u_R + \frac{p_L - p_R}{\rho c}\Big) \end{pmatrix} \tag{2.43}$$

2.3.3.2. Riemann Solvers for non-linear systems

As we know a more complicated case arises when the characteristic speeds depend on $\boldsymbol{u}$ resulting in non-linear relationships between the state $\boldsymbol{u}$ and the flux. As usual let's start with the scalar case with convex or concave flux. In this case we only have one characteristic that is genuinely non linear. The shock case is solved using the Rankine Hugoniot condition and the rarefaction one using the self similarity solution.

Example 2.3.3.2.1 (Burgers' equation):

The Burgers' equatio, given by (2.4), is a non linear scalar conservation law with a convex flux.

When $u_L > u_R$, $\lambda(u_L) > \lambda(u_R)$ and we have a shock. To compute its speed and direction, we use the *Rankine-Hugoniot Condition* (2.14):

$$\frac{1}{2}(u_R^2 - u_L^2) = S(u_R - u_L) \tag{2.44}$$

and we obtain the shock wave speed as the mean between the left and right state:

$$S = \frac{1}{2}(u_R + u_L) \tag{2.45}$$

If $S > 0$ then the shock is moving to the right and if $S < 0$ then the shock is moving to the left. This gives us the solution :

$$u(x,t) = \begin{cases} u_L \text{ if } x - St < 0 \\ u_R \text{ if } x - St > 0 \end{cases} \tag{2.46}$$

Now for the case $u_L < u_R$ we get a rarefaction as $\lambda(u_L) < \lambda(u_R)$. The speed of the head and tail are given by the characteristic speeds of the left and right states $\lambda(u_L) = u_L$ and $\lambda(u_R) = u_R$. As λ is monotone, it can be inverted and from the self similarity solution gives $u(x,t) = \lambda^{-1}\left(\frac{x}{t}\right) = \frac{x}{t}$ inside the rarefaction. Finally the solution to the Riemann problem is given by :

$$u(x,t) = \begin{cases} u_L \text{ if } \frac{x}{t} < u_L \\ \frac{x}{t} \text{ if } u_L < \frac{x}{t} < u_R \\ u_R \text{ if } \frac{x}{t} > u_R \end{cases} \tag{2.47}$$

Sampling the result at $u(0,t)$ directly gives us the godunov flux as $f(u(0,t)) = u(0,t)$.

Solving the Riemann problem for non-linear hyperbolic system on the other hand gets more complicated as we have m waves that can't be decoupled from one another like in the linear case. Instead we start from leftmost state and use the properties of the different waves we can encounter, like the Rankine Hugoniot conditions for shocks and the Generalized Riemann invariants across rarefactions. Each wave supplies $m-1$ relations and therefore leaves one free parameter. So waliking across the m waves leaves m unknowns that we can close by imposing the last state to equal the $\boldsymbol{u}_R$ giving m conditions. For the Euler equations this reduces to a single scalar equation, solved by a root-finding method as detailed in Section 2.3.3.3.

For more details look at the end of chapter 2 and chapter 3 of the book by Toro [2] (pp. 76-85, pp. 115-138).

2.3.3.3. The exact Riemann Solver for the Euler equations

As seen when stuyding the Euler equations, the solution to the Riemann problem has 4 regions, on the left side of the left non-linear wave, we have $\boldsymbol{u}_L$, then on the right side of the right nonlinear wave, we have $\boldsymbol{u}_R$. The region in between the two non linear waves is called the *star region*, it is split in two by a contact wave. Pressure and particle velocity are constant in the star region but density changes accross the contact wave. We thus have to determine the 4 unknowns ρ_L^*, ρ_R^*, u^* and p^* as well as the left and right non linear waves (shock speed and the solution inside the rarefaction fans).

The first step of the exact riemann solver is to determine u^* and p^*. For this we distinguish the cases of shocks and rarefactions on the left and right sides. For shock waves we start by changing the reference frame to align with the shock speed. This results in the shock speed in the reference frame to be null. We can than apply the Rankine-Hugoniot conditions that are simplified :

$$\boldsymbol{F}(\hat{\boldsymbol{u}}_{\boldsymbol{K}}) = \boldsymbol{F}(\hat{\boldsymbol{u}}^*), \tag{2.48}$$

with $\hat{\boldsymbol{u}}$ the conserved variables in the new frame, where only the particle velocities are adapted and K being either left or right. From this we can deduce the relations

$$\begin{cases} u^* = u_L - f_L(p^*, \boldsymbol{w}_L) & \text{for the left side case} \\ u^* = u_R + f_R(p^*, \boldsymbol{w}_R) & \text{for the right side case} \end{cases} \tag{2.49}$$

where

$$f_K(p^*, \boldsymbol{w}_K) = (p^* - p_K)\sqrt{\frac{2}{((\gamma+1)\rho_K)\left(p^* + \frac{\gamma-1}{\gamma+1}p_K\right)}}, \tag{2.50}$$

is a function that only depends on p^* and the known state $\boldsymbol{w}_K$ (left or right).

For the Rarefaction cases, we also obtain the same relations as (2.49) but with a different function f_K. Using the isentropic law and the generalised Riemann invariants, one can obtain :

$$f_K(p^*) = \frac{2a_K}{\gamma-1}\left[\left(\frac{p^*}{p_K}\right)^{\frac{\gamma-1}{2\gamma}} - 1\right]. \tag{2.51}$$

We obtain two equation of the form (2.49), one for the left non-linear wave and another for the right. Subtracting them yields

$$f_L(p^*, \boldsymbol{w_L}) + f_R(p^*, \boldsymbol{w_R}) + u_R - u_L = 0. \tag{2.52}$$

This means that p^* is the root of the function :

$$f(p^*, \boldsymbol{w_L}, \boldsymbol{w_R}) = f_L(p^*, \boldsymbol{w_L}) + f_R(p^*, \boldsymbol{w_R}) + u_R - u_L \tag{2.53}$$

That can be found using Newton's method as it is differentiable and has a simple behaviour. We can then determine u^* using (2.49). This leaves to determine the densities while determining the non-linear waves. The type of non-linear wave is determined by the difference between the pressure in the star region and the outside pressure. When the pressure in the star region is greater to the exterior one we have a shock. We get the density of the left/right star region from

$$\rho_K^* = \rho_K \left[\frac{\frac{p^*}{p_K} + \frac{\gamma-1}{\gamma+1}}{\frac{\gamma-1}{\gamma+1}\frac{p^*}{p_K} + 1} \right] \tag{2.54}$$

and the shock speed

$$S_L = u_L - a_L \sqrt{\frac{\gamma+1}{2\gamma}\frac{p^*}{p_L} + \frac{\gamma-1}{2\gamma}} \tag{2.55}$$

or

$$S_R = u_R + a_R \sqrt{\frac{\gamma+1}{2\gamma}\frac{p^*}{p_R} + \frac{\gamma-1}{2\gamma}} \tag{2.56}$$

When the pressure in the star region is smaller than the exterior one, we have a rarefaction wave. We get the density from the isentropic law

$$\rho_K^* = \rho_K \left(\frac{p^*}{p_K} \right)^{\frac{1}{\gamma}} \tag{2.57}$$

and the sound speed is

$$a_K^* = a_K \left(\frac{p^*}{p_K} \right)^{\frac{\gamma-1}{2\gamma}} \tag{2.58}$$

For a left rarefaction wave, the speed of the head and tail are given by

$$\begin{aligned} S_{HL} &= u_L - a_L \text{ for the head} \\ S_{TL} &= u_L^* - a_L^* \text{ for the tail} \end{aligned} \tag{2.59}$$

For a right rarefaction wave, the speed of the head and tail are given by

$$\begin{aligned} S_{HR} &= u_R + a_R \text{ for the head} \\ S_{TR} &= u_R^* + a_R^* \text{ for the tail} \end{aligned} \tag{2.60}$$

The solution inside a rarefaction fan can be found using the Generalised Riemann Invariants and the slope of the characteristic emanating from the origin to the point (x, t) inside the fan.

Finally we can sample the solution at $(0, t)$ and compute the flux for our finite volume solver.

This solver is exact but may not be computationally friendly for big problems as we have to solve a root finding problem. Furthermore it doesn't allow for vacuum (division by zero) which can happen when removing all the density of a cell.

2.3.4. Approximate Riemann Solvers for the Euler system

In order to reduce the computational complexity of the godunov's method, many approximate Riemann solvers have been developed. The first called *approximate state Riemann Solvers* give an estimate of $\boldsymbol{u}^*(0, t)$ and then compute the flux normally. Others like HLL, HLLC and Roe's solvers approximate the flux $F(\boldsymbol{u}^*(0, t))$ directly.

2.3.4.1. Approximate state RS

The first type of approximation done is replacing the exact value of p^* found by Newton method by a closed form estimate. The rest of the algorithm stays the same. The primitive variable Riemann solver (PVRS), linearises the primitive variable form of the Euler equations about the arithmetic mean. This gives $p^* = \frac{1}{2}(p_L + p_R) - \frac{1}{2}(u_R - u_L)\bar{\rho}\bar{a}$, see [2] p299. Another possibility is to assume that both non linear waves are of the same type (either two shocks or two rarefactions) which makes it possible to determine p^* without finding a root. These are called the two-rarefaction Riemann solver (TRRS) and the two-shocks Riemann solver (TSRS).

These solvers are useful to accelerate the Godunov method on parts where the flow is simple. Then using the exact Riemann solver only on large gradients and approximate state Riemann solver on smooth parts of the flow result in *Adaptive Riemann solvers.*

2.3.4.2. HLL and HLLC

The HLL solver assumes that there is no middle contact wave and thus there is a single unknown state $\boldsymbol{u}^{\mathrm{HLL}}$. This makes it possible to directly compute the flux, see [2] chapter 10. It is positivity preserving and is entropy satisfying by construction making it very robust. However, removing the contact wave creates excessive dissipation smearing contacts and shear waves. An improvement is to restore a middle contact wave which results in the HLLC method. Both of these methods approximate the two or three waves speeds and with the usual condition on each wave this gives a closed form solution.

2.3.4.3. Roe's solver

The idea of Roe's solver is to linearize the problem and use the exact linear Riemannn solver which is much simpler. This means replacing the jacobian of the flux $A(\boldsymbol{u})$ by a constant matrix $\tilde{A}(\boldsymbol{u}_L, \boldsymbol{u}_R)$ that fulfills the three key properties :

- $\tilde{A}(\boldsymbol{u}, \boldsymbol{u}) = A(\boldsymbol{u})$
- $\tilde{A}(\boldsymbol{u}_L, \boldsymbol{u}_R)$ should be hyperbolic
- $\tilde{A}(\boldsymbol{u}_L - \boldsymbol{u}_R) = F(\boldsymbol{u}_R) - F(\boldsymbol{u}_L)$

For the Euler equation it is obtained by evaluating the jacobian at the Roe averages, see [2] chapter 11.

As the only wve types in linear systems are discontinuities, this means that Roe's solver represents rarefactions with a discontinuity. This can lead to expansion shocks that don't fulfill Lax's entropy condition. It thus requires an *entropy fix*.

3. Implementation in Julia

In order to truely understand the computational side of Godunov's method, I implemented it in Julia. It is a high level programming language designed for scientific computing, ideal for quick implementations that require performant code.

3.1. Structure of the code

The dependencies of the Julia project are defined in the `Project.toml` file and they are versionned in the `Manifest.toml` file. This makes it easy to use the same packages on other devices.

3.1.1. Source code

The source code is divided in multiple folders, each defining a specific part of the solver. The custom **mesh** structure holds the data of the mesh (points, cells, faces, boundaries) as well as some precomputed values used in the solver loop. Some helper functions makes it easy to generate 1D regular meshes and load 2D meshes from a `.msh` file. The solver uses types to distinguish between different **equations**. To define an equation one needs to define the number of variables, a numerical flux and a cfl condition. **Boundary conditions** are also defined in custom structures and are solved using ghost cells. The solver interface then requires a mesh, an equation type and a dictionnary linking physical borders to boundary conditions. The solver in itself uses the finite volume method as in Section 2.3.1. The data is written in `.vtu` files during the simulation and can then be visualized with dedicated softwares like Paraview. The data to be written is defined by a method `output_fields` for each type of equation.

3.2. Models

Some models are already implemented and can be directly used. We started with linear models like the linear transport and the wave system. For such equations we proceed as described in Section 2.3.3.1 and compute the numerical flux from the upwind direction. Then we experienced with the non linear Burgers' equation. To compute the numerical flux we proceeded as in Example 11. In comparison to the linear case, we now had to compute the fastest wave dynamically as it is state dependent. Finally we implemented the full Euler equations for ideal gases using the Godunov method with multiple Riemann solvers.

3.3. Results

3.3.1. Linear Advection

The linear advection problem is very simple as the initial and border conditions are simply advected following the vector $\boldsymbol{c}$. As such we can compare the result from the finite volume method to the exact solution.

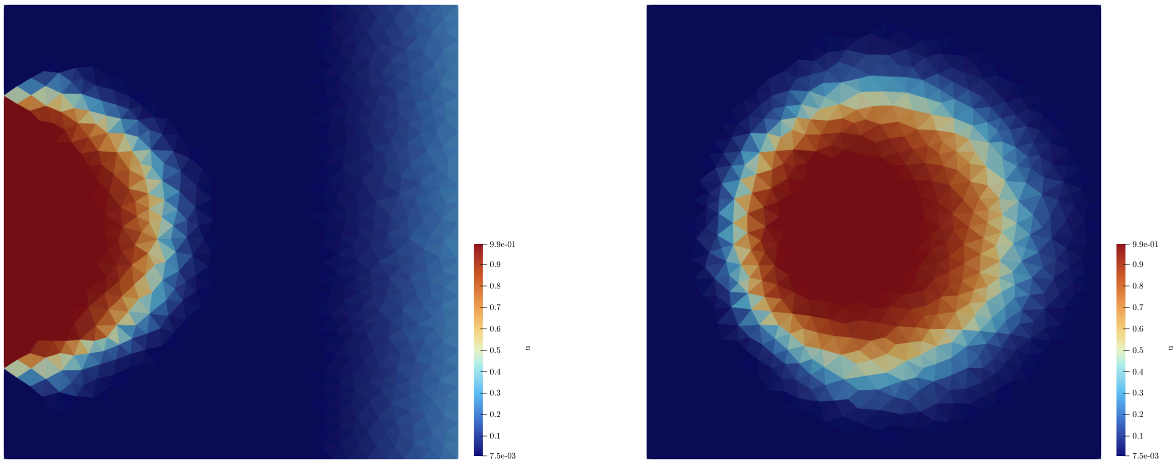

Figure 7: Linear Advection in 2D, a disk being advected from left to right

This example shows a disk defined on the boundary condition being linearly advected to the right. The exact solution would be a perfect round circle with value 1. We see that the method is diffusive as the circular shape isn't perfectly clean. Furthermore the front is more diffused than the rear as it has been transported for longer.

3.3.2. Euler equations

For the Euler equations we implemented 4 numerical fluxes, the Godunov flux with an exact Riemann solver as well as the HLL, HLLC and the Roe flux. To look at something more aerodynamic than a simple circle, we used the NACA0012 airfoil exposed to a Mach 3 flow. Its contour can be easily defined by a curve as given on Wikipedia.

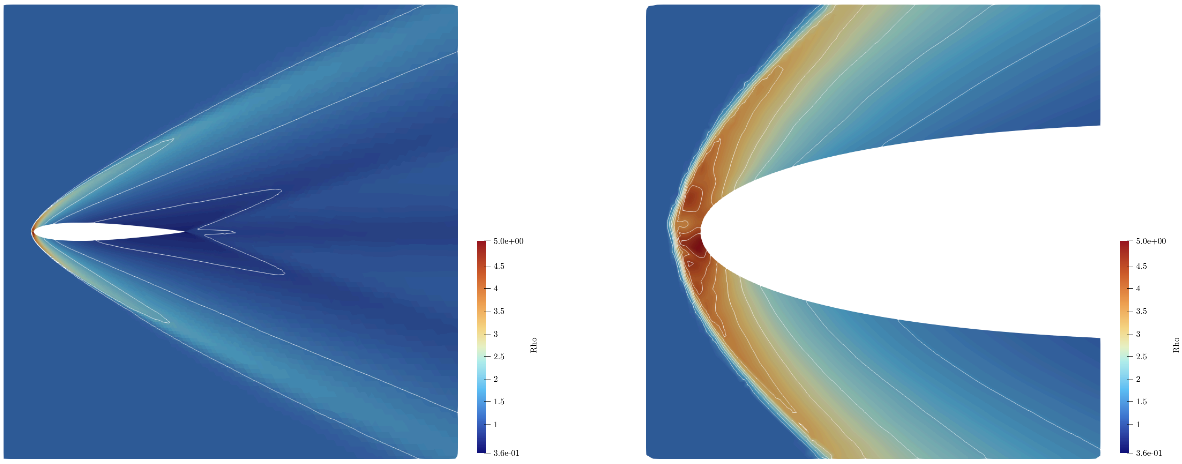

Figure 8: Density plot with isolines of a NACA0012 airfoil at Mach 3 (exact Riemann solver).

On the front of the airfoil we have a shock as we can see a rapid jump in the state values and packed isolines. When zooming on the front of the airfoil we can see some irregularities in the density field. A more severe case can arise with particular conditions. By increasing the flow speed and using an object that is more blunt and a grid that aligns with the shock a carbuncle like disruption is created on the shock.

3.3.3. Performances

Performance wise the implementation seems to be on par with high performance solvers like Aerosol. It is still only single threaded, but adding multiprocessing and gpu can be done without changing to much of the code. The only part that would need rework is the main finite volume loop inside the solver such that it doesn't break race conditions.

4. Numerical Instabilities in Godunov-type methods

Usually complete Riemann solvers, like exact Riemann solvers, Roe's solver and HLLC are considered better in the sense that they capture shocks more sharply than uncomplete solvers like HLL would. In 1994, J.Quirk published a paper [4] raising awareness on flaws observed on Godunov-type methods. Complete Riemann solvers can create spurious solutions that are not observed in more diffusive solvers. For a study of the origin of the instability and possible cures see [5].

4.1. Test cases

In order to check if a numerical scheme is subject to the carbuncle, one has to go through a lot of tests as described in [6]. The main test cases were implemented in the Aerosol software for the isentropic Euler equation with $\gamma = 2$ and $\kappa = 0.5$.

4.1.1. Shock travelling a duct

The first testcase is simply checking the behaviour of a shock travelling through a duct. The domain is a 800×20 rectangle with a structured mesh.We set a shock wave with Mach 6 at the start of the duct and let it travel to the end of the domain.

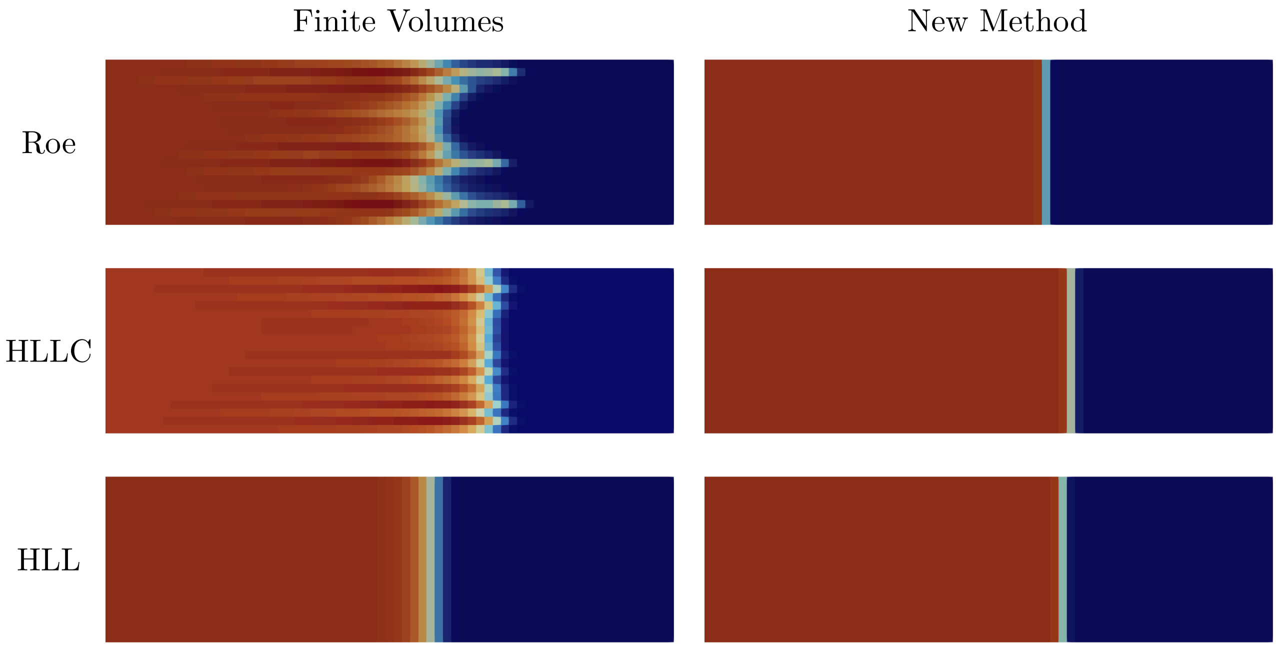

Figure 9: Mach 6 shockwave at the end of a 800×20 duct

As expected the HLL scheme smears the shock wave and doesn't develop any instabilities for both method. On the other hand using complete Riemann solvers results in the frontal shock being destroyed into a spike like structure for the standard finite volumes method. This isn't the case for the curl and divergence preserving method where the shock stays sharp.

4.1.2. Odd-even decoupling

The instabilities seen in the previous case come from rounding errors in the mesh alignement. In order to accentuate and control these instabilities we reproduced the Odd-Even decoupling example introduced by Quirk in [4].

The same initial condition and domain are used but we include a more sensible alignement error in the middle points of the mesh, creating a zig-zag:

$$\begin{aligned} Y_{i,j\ \text{mid}} &= Y_{j\ \text{mid}} + 10^{-6} \text{ for } i \text{ odd} \\ Y_{i,j\ \text{mid}} &= Y_{j\ \text{mid}} - 10^{-6} \text{ for } i \text{ even} \end{aligned} \tag{4.61}$$

This makes it possible to watch the previous spikes grow earlier in a more regular and controlled way.

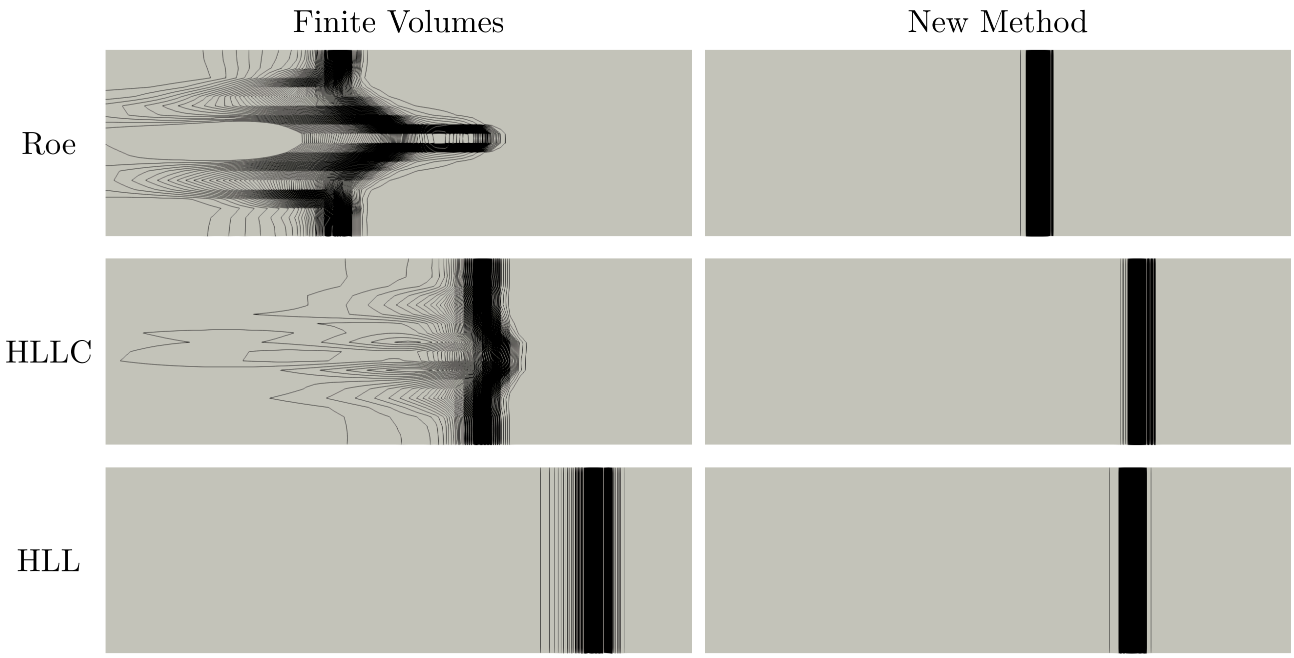

Figure 10: Contour lines at $X_s \approx 480$ of the momentum magnitude of a Mach 6 shock travelling a duct with odd even decoupled instabilities in the mesh

We can observe the standard finite volume methods behaving like in Quirk's paper [4], the complete Riemann solvers leading to instabilities and the incomplete HLL solver's shock being diffused. The new method on the other hand keeps a sharp shock for every Riemann solver.

4.1.3. Supersonic flow around blunt body

The following test case is the reason the studied instability is called the *Carbuncle instability*. Placing a blunt body inside of a supersonic flow generates a shock in front of its nose that can destabilizes into a sort of carbuncle. This is especially observed when aligning the grid with the shock front.

The domain consist of the area between two half-circles filled with a regular mesh of quadrangles. The data is initialized to a given Mach flow perpendicular to the blunt body and the outer circle acts as an inflow.

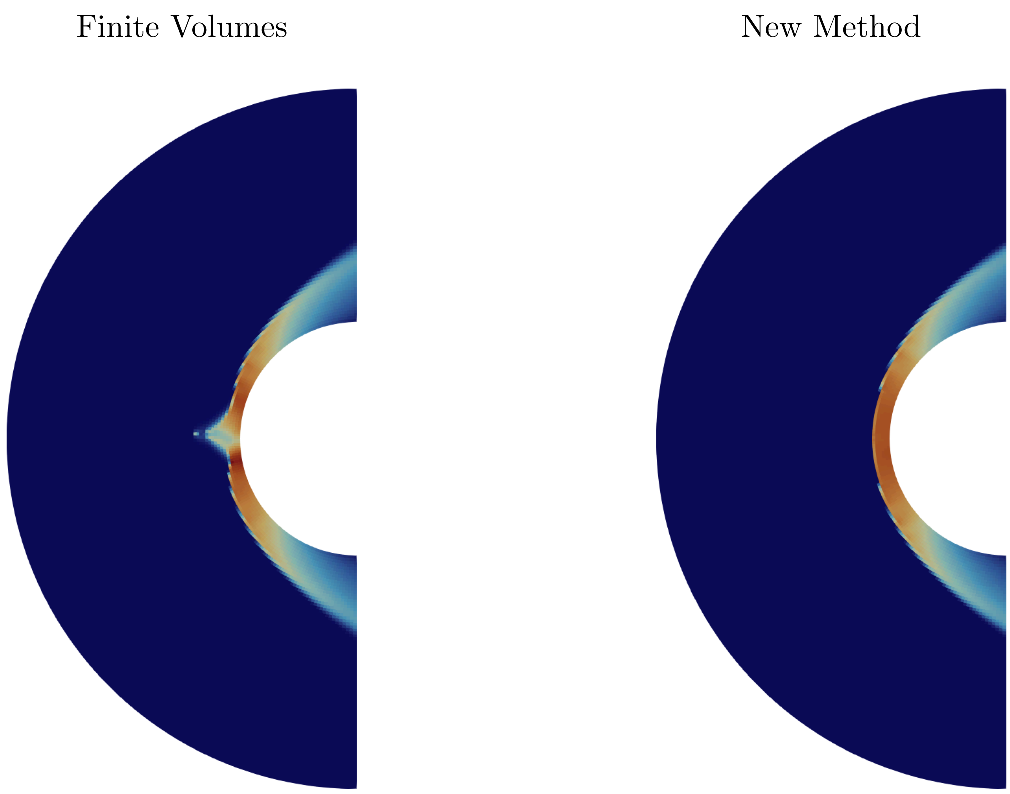

Figure 11: Density plot of the Carbuncle using the Roe solver

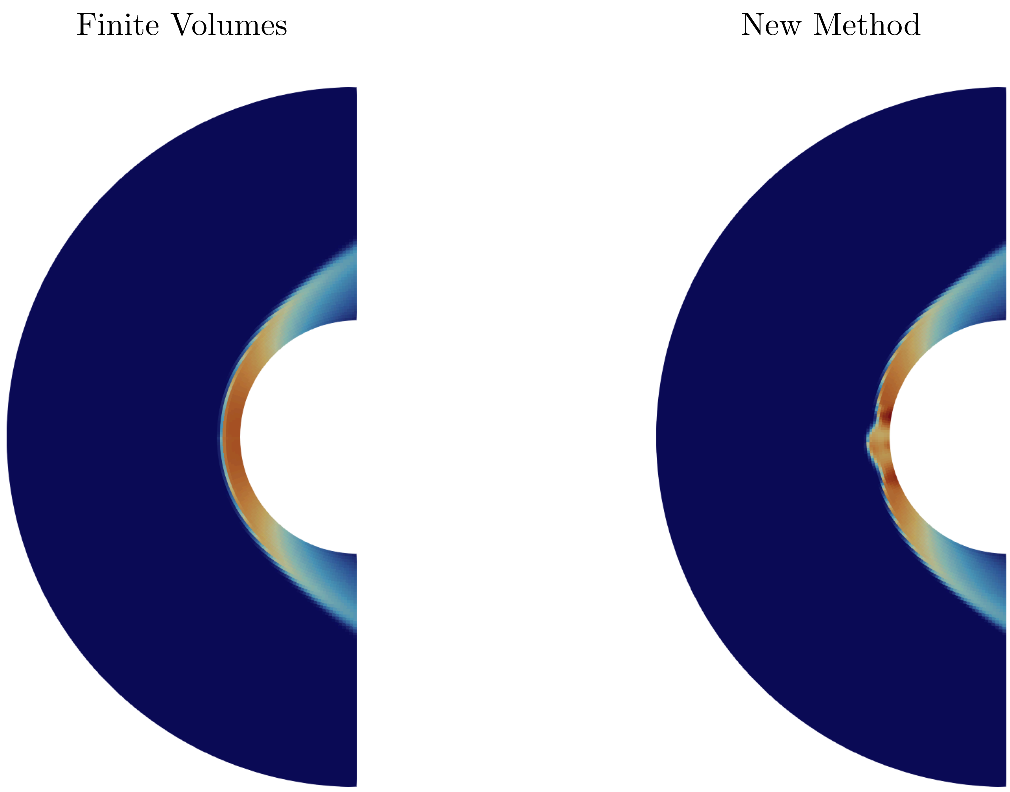

Figure 12: Density plot of the Carbuncle using the HLLC solver

For the Roe solver, we see a carbuncle instability in the Finite Volume result but the new method doesn't trigger the instability and we get a sharp shock front. On the other hand using the HLLC solver gets us the opposite results. The finite volumes don't trigger the carbuncle and the new method does. Weirdly enough using the standard finite volumes with HLLC results in a diffuse shock interface eventho it is a complete Riemann solver.

This testcase isn't complete it would be better to do multiple simulation with different Mach values, ranging from transonic to hypersonic speeds. Even if the Carbuncle didn't appear using the new method and the roe solver at Mach 10 doesn't mean it wouldn't appear at Mach 20.

4.1.4. Elling test

The Lax-Wendroff theorem tells us that conservative solvers converge to weak solutions but unicity of such weak solutions is not assured in multiple dimensions. In [7], Elling stated that these observed instabilities may be admissible entropy solutions and thus are incurable. In order to illustrate this, Elling designed a test in which the carbuncle can be observed to be "physically correct". The setting is inspired from the testcases in [8] and [9]. We place a steady Mach 30 shock at $x = 50$ on a $[0, 100] \times [0, 40]$ domain discretized by 800×320. This gives for the upstream variables $(\rho, u, v) = (1.0, 30.0, 0.0)$ and downstream variables $(\rho, u, v) = (41.929, 0.71549, 0.0)$. Then we immitate a filament in the upstream part by setting $u = 0.0$ for $19.75 < x < 20.25$. The final time is set to $t = 1.0$.

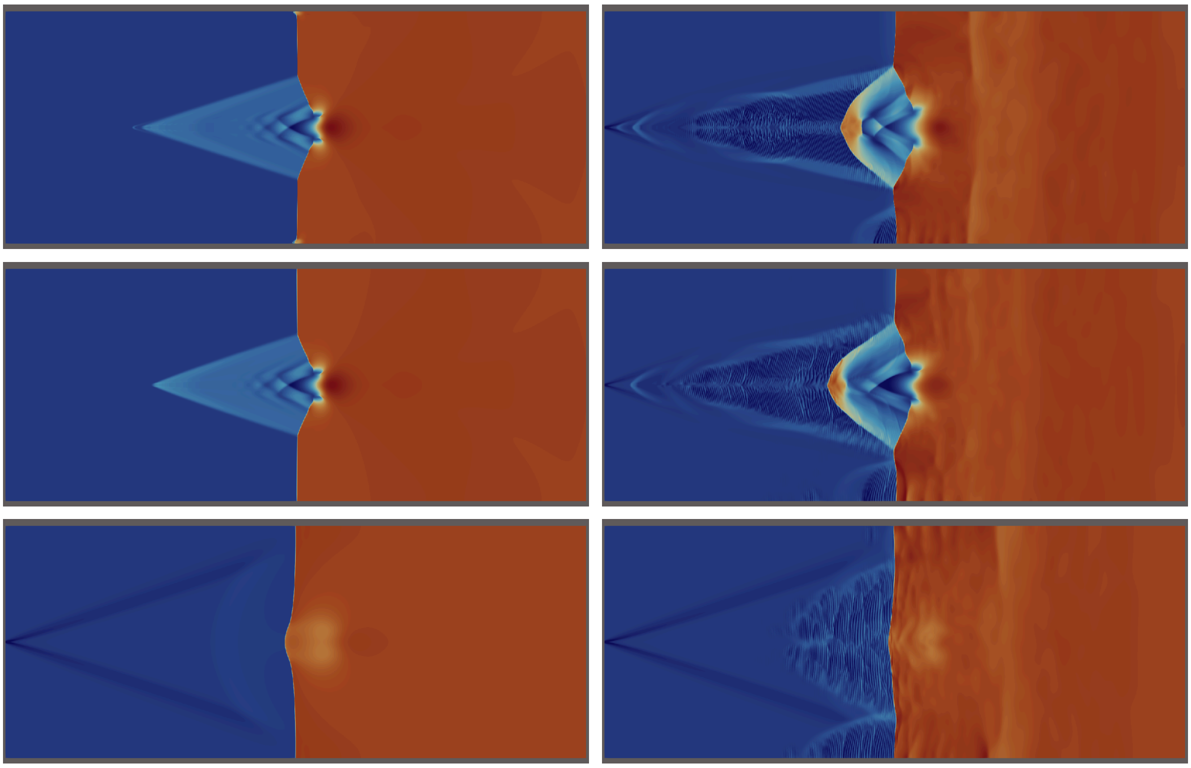

Figure 13: Density plot of the Elling test for the isentropic Euler Equation ($\kappa = 0.5, \gamma = 2$)

The complete Riemann solver create sharp carbuncle like structure and the uncomplette HLL solver supresses using both methods. The new curl and divergence preserving method creates spurious oscilation both in the upstream and downstream regions that is not observed in the classical finite volumes.

4.1.5. Kelvin-Helmholtz instability

Finally as an additional experience I wanted to test the sensitivity of the new method to the KH instability, originating from a perturbation in a shear wave. The instability grows into vortexes that are expected to be sharper using the new method as it is preserving the curl.

The domain is $[0, 2] \times [0, 1]$ discretized by a 300×150 regular mesh. We set the left and right boundaries to be periodic and the top and bottom to a null gradient outflow. Then we divide the domain into a left and right moving fluids with a smooth transition layer defined by the function H as described in [6].

$$H(y) = \begin{cases} -\sin\left(\pi \frac{y+\frac{1}{4}}{\omega}\right) & \text{if} -\frac{1}{4} - \frac{\omega}{2} \leq y < -\frac{1}{4} + \frac{\omega}{2} \\ -1 & \text{if} -\frac{1}{4} + \frac{\omega}{2} \leq y < \quad \frac{1}{4} - \frac{\omega}{2} \\ \sin\left(\pi \frac{y-\frac{1}{4}}{\omega}\right) & \text{if} \quad \frac{1}{4} - \frac{\omega}{2} \leq y < \quad \frac{1}{4} + \frac{\omega}{2} \\ 1 & \text{otherwise} \end{cases} \tag{4.62}$$

Furthermore a vertical perturbation is also added to trigger the instability.

$$(\rho, u, v) = (\gamma + rH(y), M_a H(y), \delta M_a \sin(3\pi x)) \tag{4.63}$$

with $M_a = 0.1$.

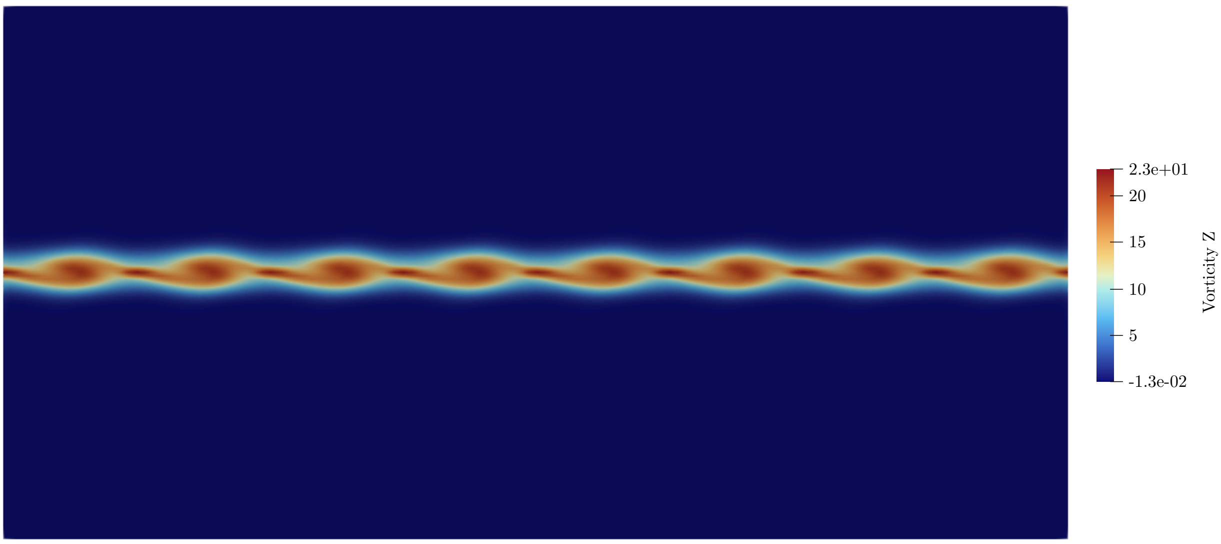

Figure 14: KH instability after 3 timesteps using the new Method with the Roe solver

Only the complete solvers generate the instability in this setting. The new methods is sensible enough to generate 8 waves in the first few steps whereas the finite volume methods only show bigger whorls that take much more time to show. These whorls then combine until a single one fills the whole domain for all complete solvers.

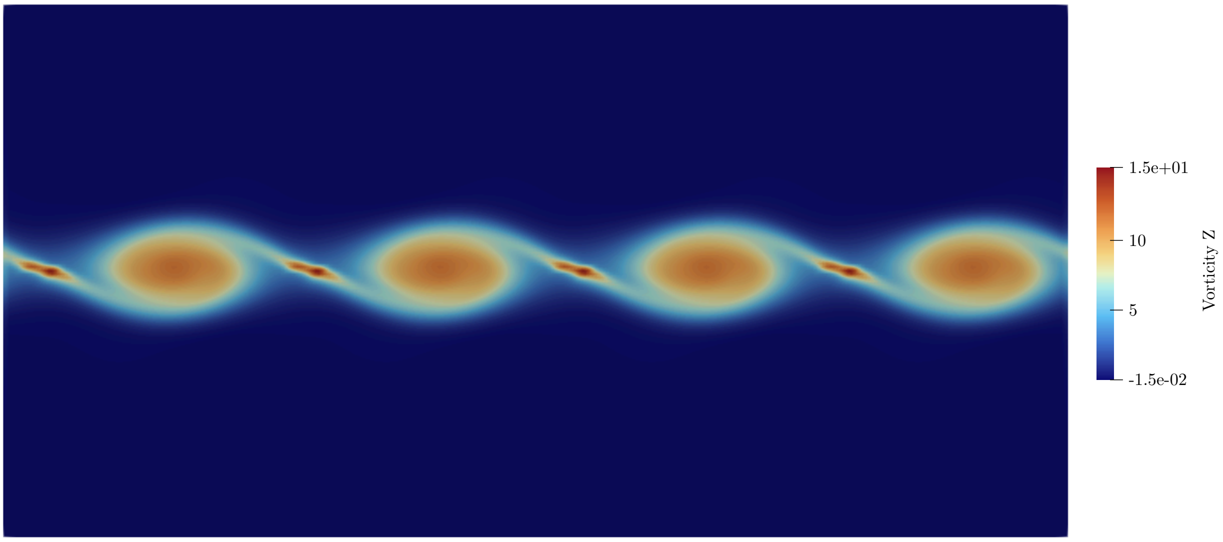

Figure 15: KH instability after 7 timesteps using the new method with the Roe solver

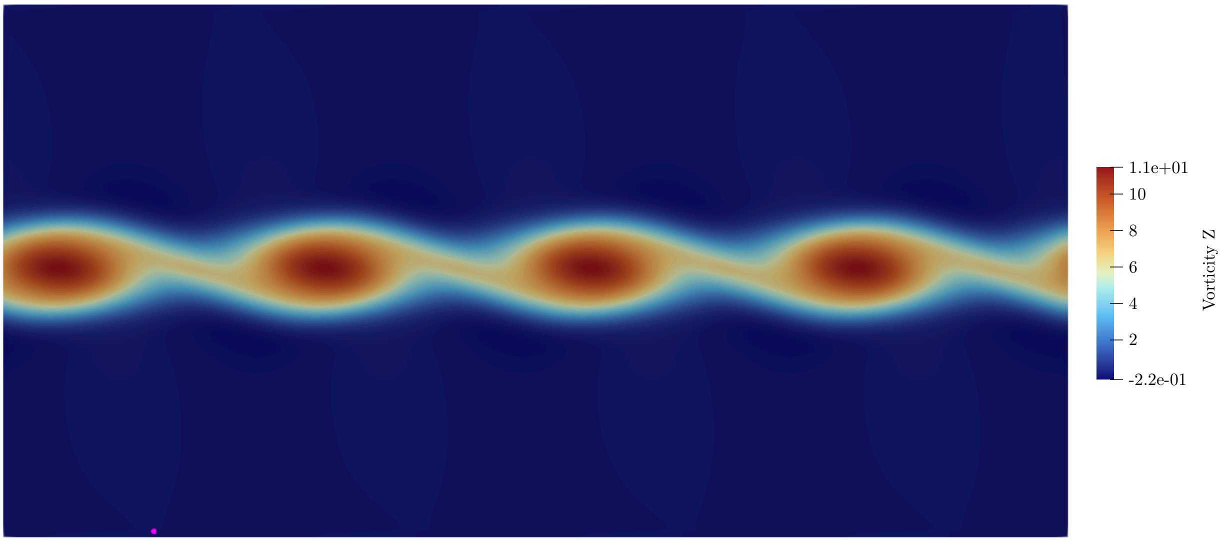

Figure 16: KH instability after 8 timesteps using finite volumes with the Roe solver

Using the HLL solver doesn't trigger the instability in all the cases, even when refining the mesh by 16 giving a 1200×600 grid.

4.1.6. Other test

There are a lot of other test I didn't get to implement. In Quirk's paper, he also included two other testcases : an expansion shock that is created when a shock encounters a stepdown, and a shock encountering a ramp that can create a kink. In [6] they mention further test done in the litterature to check grid-aligned instabilities.

5. Conclusion

All in all this internship let me study the Godunov method and Riemann solver deeply both by writting its description in this report and implementing it in the Julia language. This made it easier to understand the problems arising from the Carbuncle phenomenon and the test cases. It also gave me different expressions of the physics and maths, once through the exercices in Toro's book, then while writing the Julia code and finally during the interpretation of the results from Aerosol.

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